

Autodesk® Scaleform®

APIs

Scaleform 4.3

Scaleform C++

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2.2

2012 6 21

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Autodesk® Scaleform® 4.3

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1	Scaleform	1
1.1	Flash	1
1.1.1		2
1.1.2		3
1.1.3		4
1.1.4		5
1.2		6
2	1	8
2.1		8
2.1.1		9
2.1.2		11
2.2	gfxfontlib.swf	11
3	2	13
3.1		13
3.1.1		15
3.1.2	Scaleform Player	16
3.1.3		17
3.1.4		18
3.2		20
4	3	21
4.1		21
4.2	Gfx::FontMap	23
4.3	Gfx::FontLib	24
4.4	Gfx::FontProviderWin32	24
4.4.1		25
4.4.2		26
4.5	Gfx::FontProviderFT2	27
4.5.1	FreeType	28

1

Scaleform

Scaleform

HTML

GFx/SWF

FreeType 2

Scaleform

Flash® Studio

SWF

Flash Studio

Flash

Scaleform

Gfx::Translator

Gfx::FontLib

FreeType2

Scaleform

Flash Studio Scaleform

Flash

Scaleform

Flash Scaleform

- Flash

Flash

Flash

-

Flash

1.1 Flash

Flash Studio

Studio

Flash

Scaleform

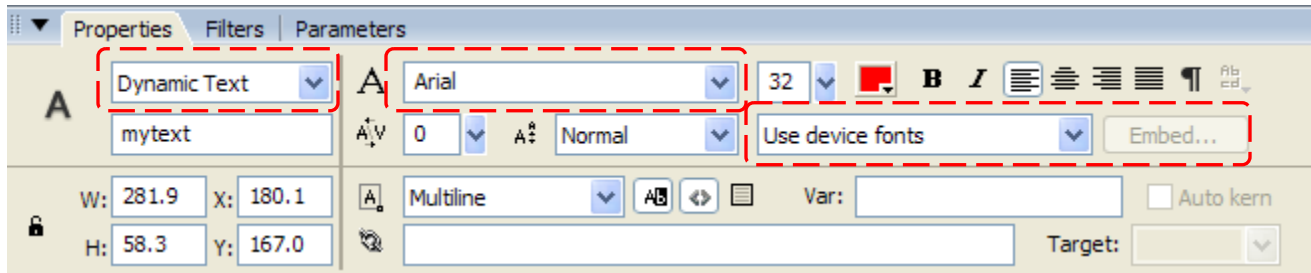
1 ---

1.1.1

Flash

Text Tool

Flash Studio



-
-
-

GFx::Translator

(a)

Arial

(b)

SWF

Flash 8

-
-

-
-

Use device fonts

SWF
Flash
/

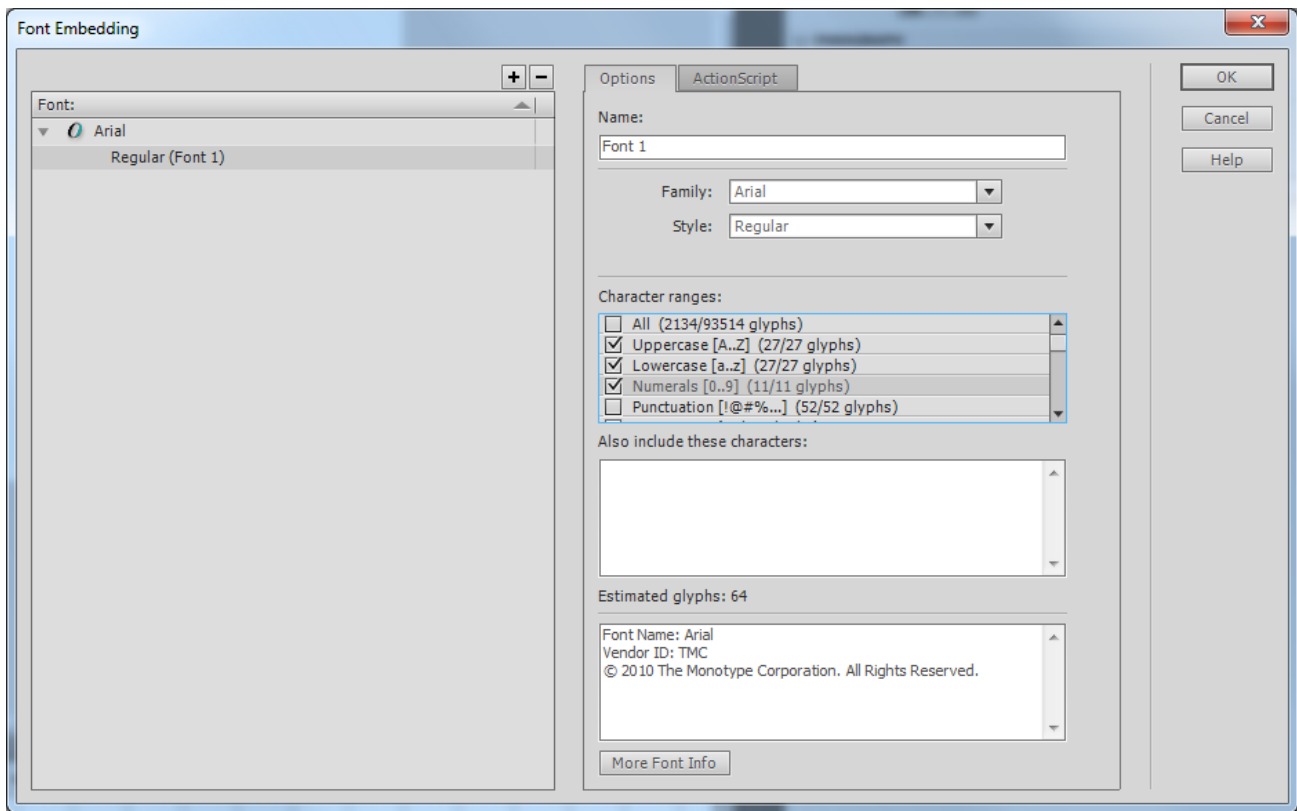
Scaleform

1.1.2

TextField

Scaleform
Embed

Scaleform



Style

Arial

Arial

Flash Studio

SWF/GFx

Scaleform
Linux

Xbox 360 PS3 Wii

Windows Mac

1.1.3

Scaleform ,

		SWF	Scaleform
1		1 KB	450 K
114	– +	12 KB	480 K
596	–	70 KB	630 K
7,583	–	2,089 KB	3,500 K
18,437	–	5,131 KB	8,000 K

Arial Unicode MS
500 150K

1.1.4

1. Flash

2. Scaleform /
SWF/GFx
Scaleform

3. IME

4. GfxExport -fc

Scaleform

IME

1.2

Scaleform

- 1 —
- 2
- 3
- 4 -

Scaleform

Scaleform

2

- Scaleform
-
-

Flash

SWF/GFx

Scaleform
3

Scaleform

Scaleform
FreeType 2

Scaleform

Scaleform
gfxexport

2 1

1.

2.

3.

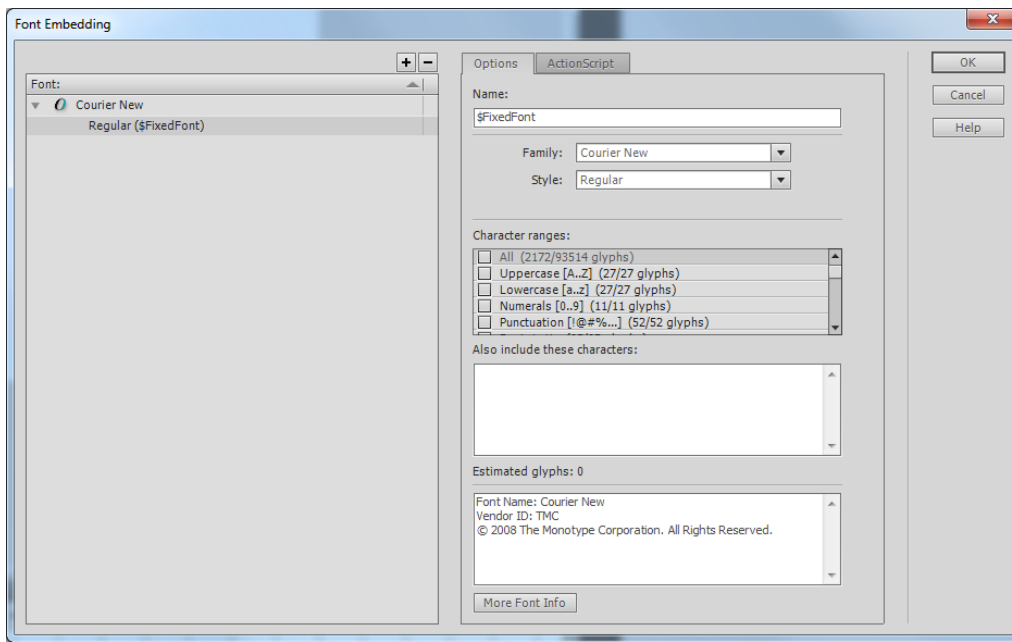
Scaleform

"fonts_en.swf" "fonts_kr.swf"
Flash

2.1

Flash Studio

New Font...



FLA

\$FixedFont

\$ArialGame*

\$

Font rendering method

Flash

Studio

2.1.1

FLA

FLA
FLA

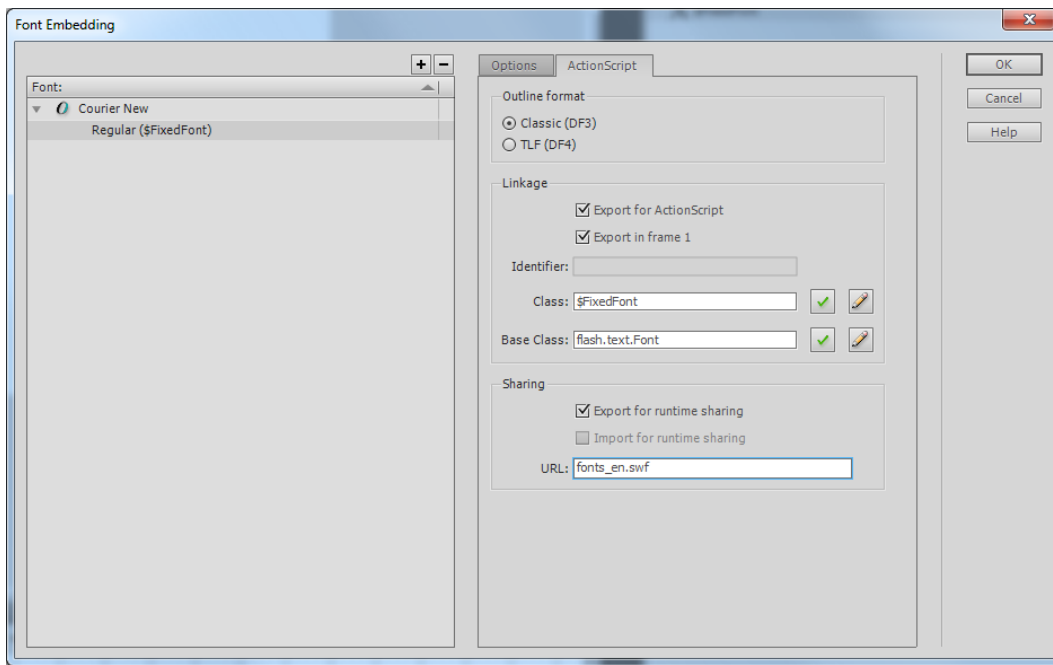
/
Copy
Paste

FLA

Flash /

"Properties"

"ActionScript"



"OK"

Flash

"Export for ActionScript"

ActionScript

"Export for runtime sharing"

"Export in first frame"

URL

SWF

GfX::FontLib

URL

"fonts_en.swf"

GfX

```
Loader.SetDefaultFontLibName("fonts_en.swf");
```

fonconfig.txt GfXPlayer

```
fontlib "fonts_en.swf"
```

fontlib

Export: \$identifier

/

SWF

Scaleform

SWF

TextField.htmlText

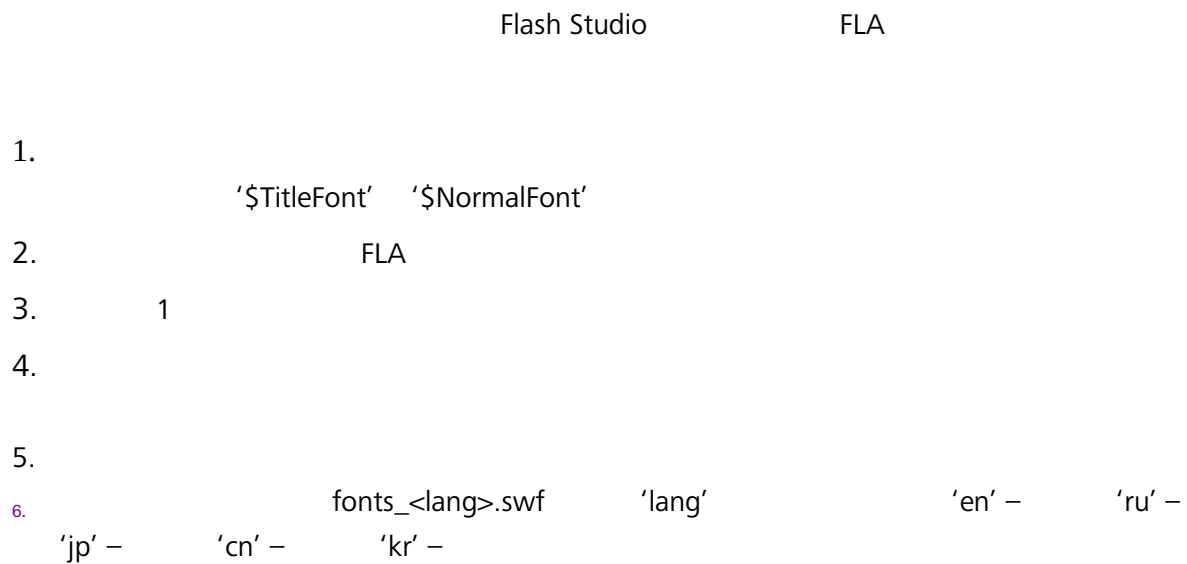
TextFormat

Arial

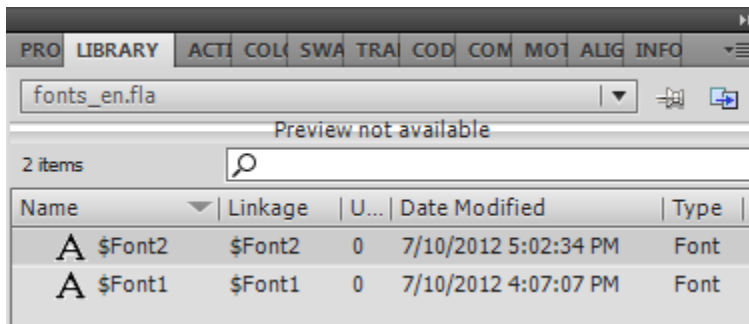
2.1.2



2.2 *gfxfontlib.swf*



FLA



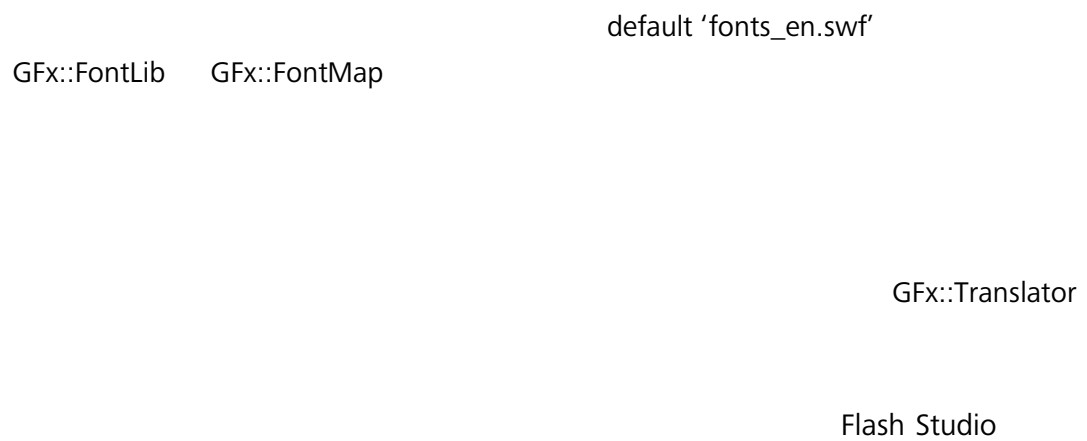
The screenshot shows the Adobe Flash Library panel. At the top, there are tabs: PRO, LIBRARY (selected), ACT, COL, SWA, TRA, COD, COM, MOT, ALIG, and INFO. Below the tabs is a text field containing 'fonts_en fla' with a dropdown arrow and two icons. A message 'Preview not available' is displayed. Below this, it says '2 items' next to a search icon. A table lists two font assets:

Name	Linkage	U...	Date Modified	Type
A \$Font2	\$Font2	0	7/10/2012 5:02:34 PM	Font
A \$Font1	\$Font1	0	7/10/2012 4:07:07 PM	Font

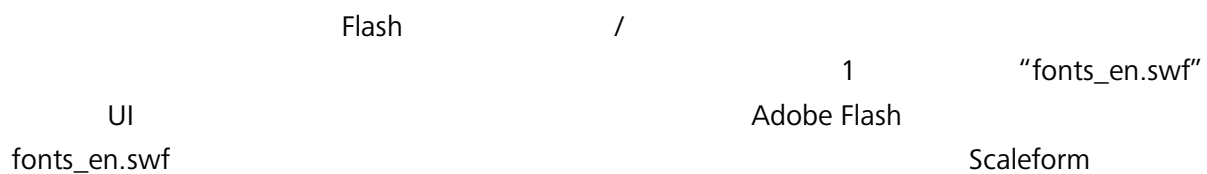
/

3 2

- 1.
- 2.
- 3.



3.1



Loader::SetDefaultFontLibName (const char filename) C++*
'filename'

```
Loader.SetDefaultFontLibName("fonts_kr.swf");
```

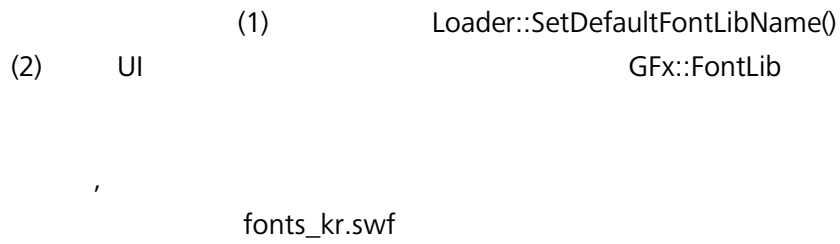
SWF CreateInstance

1.

2.

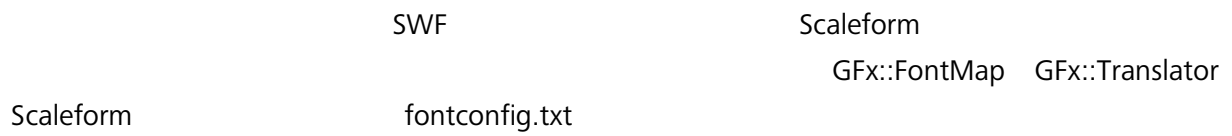
3. \$NormalFont

Scaleform	Gfx::Translator	Gfx::FontLib	Gfx::FontMap
Gfx::Loader			Gfx::Loader::CreateMovie



UIScreen

3.1.1



1. 1 "fonts_en.swf"
2. Gfx::Translator
3. Gfx::FontMap
4. "fonts_kr.swf"

1. Gfx::Loader Gfx::Translator
2. Gfx::FontMap 3 Gfx::Loader MapFont
Gfx::FontMap
3. Gfx::FontProvider /
Gfx::FontPackParams GlyphCacheConfig
4. Gfx::FontLib Gfx::Loader

5. swf SetDefaultFontLibName(filename)

Loader.SetDefaultFontLibName("fonts_en.swf");

Gfx::FontLib SWF/GFx
Gfx::FontLib::AddFromFile

6. Gfx::Loader::CreateMovie

3.1.2 Scaleform Player

Scaleform Player fontconfig.txt
SWF/GFx 'fontconfig.txt'
/fc Ctrl+N F2
HUD
8 ASCII UTF-16

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$TitleFont" = "Times New Roman" Normal
map "$NormalFont" = "Arial " Normal
map "$SmallFont" = "Verdana" Normal
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
tr "Hello" = "안녕하세요"
tr "Thank You" = "감사합니다"
```

```
[FontConfig "Chinese"]
fontlib "fonts_ch.swf"
map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal
tr "Hello" = " "
tr "Thank You" = " "
```

[FontConfig "name"]

Scaleform Player HUD

fontlib "fontfile"	SWF/GFx GFx::FontLib fontlib
map "\$UIFont" = "PMingLiU"	GFx:FontMap GFx:FontMap \$UIFont fonts_en.swf UI
tr "Hello" = " "	

Scaleform

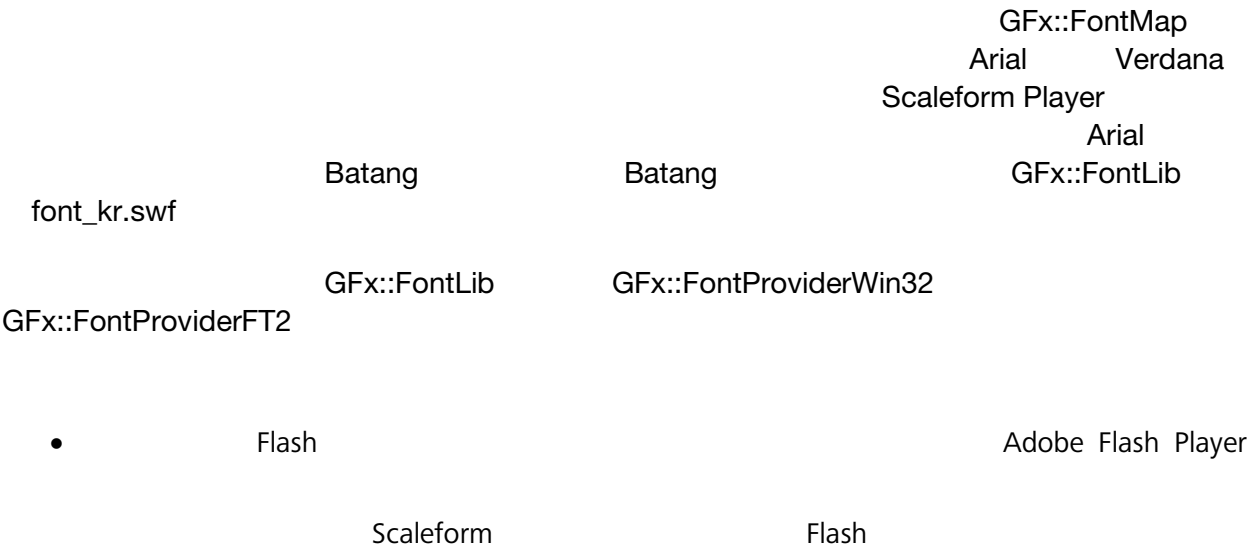
XML FontConfigParser.h/.cpp

Bin\Data\AS2\Samples\FontConfig and Bin\Data\AS3\Samples\FontConfig

Scaleform sample.swf fontconfig.txt

sample.swf Scaleform Player fontconfig.txt

3.1.3



- Anti-alias for animation ActionScript
 TextField.antiAliasType Scaleform Player
- Scaleform Adobe Flash
 ("fonts_en.swf")
- ActionScript TextFormat Flash \$TitleFont
 Arial
- fonts_en.swf
 fonts_en.swf

3.1.4

SWF SWF fontconfig.txt

FLA ActionScript 2.0 3.0 'main-app.flas'

SWF

FLA ActionScript 'fonts_en.swf'

'Library' 'New Font...' ...

\$Font1 "Family" "Arial" "Style" "Regular" English 'Basic

"Character ranges"

Latin'

'ActionScript' 'Export for ActionScript' ActionScript 'Export for

frame 1' 1 'Export for run-time sharing' 'URL'

'fonts_en.swf' 'OK' ActionScript 3.0 Flash Studio

'A definition for this class could not be found...' ... -

'OK'

3 - 5 '\$Font2' 'Family' / 'Style'

'Family' 'Style' Flash Studio

'Times New Roman'

FLA ActionScript 'fonts_kr.flas' 'fonts_ru.flas'

Korean 'fonts_kr.flas'

'Library' 'New Font...' ...

\$Font1 "Family" "바탕" "Style"

"Regular"

"Character ranges"				'Korean Hangul	
(All)	'ActionScript'	'Export for ActionScript'	ActionScript	'Export for	
frame 1'	1	'Export for run-time sharing'		'URL'	
	'fonts_en.swf'	'OK'	ActionScript 3.0	Flash Studio	
	'A definition for this class could not be found...'			...	-
	'OK'				
	8 - 10	'\$Font2'	'Family'		
'바탕체'					
	'fonts_en.swf'	'fonts_kr.swf'			
	main-app.fla			'fonts_en.swf'	
Library		'\$Font1'	'\$Font2'	Ctrl-C	->
'Copy'					
	'main-app.fla'	Library	Ctrl-V	-> 'Paste'	
		'Import:'			
'main-app.fla'			'Classic Text'		'Dynamic
Text'		'Family'	'\$Font1*'	'\$TEXT1'	
		ID			
		'\$Font2*'	'\$TEXT2'		
'main-app.swf'					
	swf		fontconfig.txt	'Notepad'	
Unicode	UTF-8				

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = "이것"
tr "$TEXT2" = "은 한국이다!"
```

	"Batang"	"BatangChe"	"바탕"	"바탕체"	Flash	SWF
				fontconfig.txt		
	SWF		gfxexport -fntlst <swfname>			

*.lst

	Unicode	UTF-8	GFXPlayer	'main-
app.swf'	'This is'	'English!'	Ctrl-N	
Korean	fonts_kr.swf		fonts_kr.swf	

3.2

	UIScreen.fl		SWF/GFx
			SWF
UIScreen.swf	UIScreen.swf		

600K

-
-
-
-

IME

		Flash Studio
	Ctrl + F11	
Replace strings	manually using stage language	Flash
Scaleform		

4 3

Flash

HTML

1.

2. SWF / GfX

3. GfX::FontLib

SWF/GfX

4. GfX::FontProviderWin32

Flash

GfX::FontLib GfX::FontMap GfX::FontProvider
GfX::FontLib GfX::FontMap SWF/GfX

GfX::Loader::SetFontProvider

SWF

SWF

Scaleform

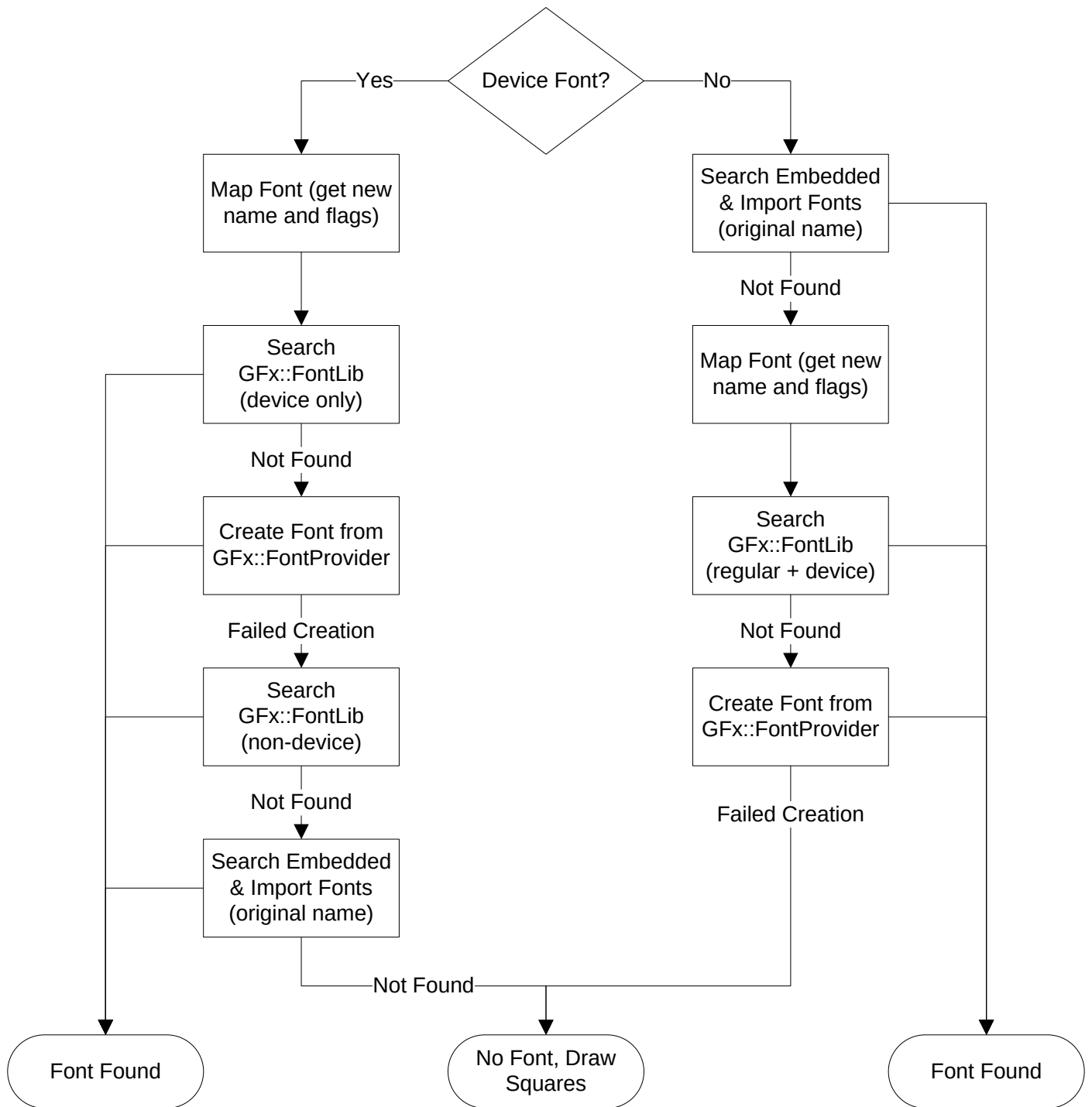
- GfX::FontProviderWin32 – Win32 APIs
- GfX::FontProviderFT2 – David Turner FreeType-2

4.1

Scaleform
Use device fonts

Adobe Flash

Scaleform



Gfx::FontLib Gfx::FontProvider

4.2 Gfx::FontMap

Gfx::FontMap
Gfx::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

\$TitleFont

C++

```
#include "Gfx/Gfx_FontLib.h"
. . .
Ptr<FontMap> pfontMap = *new FontMap;
Loader.SetFontMap(pfontMap);

pfontMap->MapFont("$TitleFont", "Batang", FontMap::MFF_Bold scaleFactor = 1.0f);
pfontMap->MapFont("$NormalFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
pfontMap->MapFont("$SmallFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
```

```
;      $NormalFont    $SmallFont
      MapFont
      MFF_Original
scaleFactor
```

scaleFactor 1.0f

Gfx::FontMap

Gfx::FontMap

Gfx::Loader::CreateMovie

Gfx::MovieDef

abcdefghijklmnopqrstuvwxyz
 ABCDEFGHIJKLMNOPQRSTUVWXYZ
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

Adobe Flash

Scaleform

Win32 FreeType

```

vectorRange = Font::DontHint;
rasterRange = Font::HintCJK;
maxVectorHintedSize = 24;
maxRasterHintedSize = 24;

```

4.4.2

SetHinting()

SetHintingAllFonts()

```
fontProvider->SetHintingAllFonts(. . .);
```

```
fontProvider->SetHinting(fontName, . . .);
```

SetHinting()

SetHintingAllFonts()

```

void SetHintingAllFonts(Font::NativeHintingRange vectorRange,
                        Font::NativeHintingRange rasterRange,
                        unsigned maxVectorHintedSize=24,
                        unsigned maxRasterHintedSize=24);

```

```

void SetHinting(const char* Name,
                Font::NativeHintingRange vectorRange,
                Font::NativeHintingRange rasterRange,
                unsigned maxVectorHintedSize=24,
                unsigned maxRasterHintedSize=24);

```

Name

UTF-8

4.5 Gfx::FontProviderFT2

David Turner FreeType-2 library
 Gfx::FontProviderWin32 *Gfx::FontProviderFT2*

FreeType

Windows MSVC

```

freetype<ver>.lib –          DLL
freetype<ver>_D.lib –        DLL
freetype<ver>MT.lib –        DLL(      CRT)
freetype<ver>MT_D.lib –      DLL (      CRT)

“<ver>”      FreeType          2.1.9      219

freetype2/include  freetype2/objs
  
```

Gfx::FontProviderFT2

```

#include "Gfx/Gfx_FontProviderFT2.h"
. . .
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);
  
```

FontProviderFT2(FT_Library lib=0);

```

FreeType                                     FreeType
      FreeType                               Scaleform
      (
FT_Done_FreeType)                           Gfx::Loader
      FreeType
fontProvider->GetFT_Library()                 FT_Library
      FreeType                               Gfx::FontProviderFT2
FreeType
  
```


4.5.1 FreeType

Win32 API

FreeType

Gfx::FontProviderFT2

```
void MapFontToFile(const char* fontName, unsigned fontFlags,
                  const char* fileName, unsigned faceIndex=0,
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                  Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                  unsigned maxVectorHintedSize=24,
                  unsigned maxRasterHintedSize=24);
```

fontName	Times New Roman	filename
C:\\WINDOWS\\Fonts\\times.ttf	fontFlags	Font::FF_Bold
Font::FF_Italic	Font::FF_BoldItalic	Font::FF_BoldItalic
Font::FF_Bold Font::FF_Italic	faceIndex	FT_New_Face
0		
;	Win32	
FreeType		

4.5.2

FreeType

```
void MapFontToMemory(const char* fontName, unsigned fontFlags,
                    const char* fontData, unsigned dataSize,
                    unsigned faceIndex=0,
                    Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                    Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                    unsigned maxVectorHintedSize=24,
                    unsigned maxRasterHintedSize=24);
```

```
FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd) {
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
}
```

```

        fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
    }

```

Gfx::Loader

FreeType

Scaleform

Windows

FreeType

```

Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
                           "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
                           "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
                           "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
                           "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
                           "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
                           "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
                           "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
                           "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
                           "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
                           "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
                           "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,
                           "C:\\WINDOWS\\Fonts\\verdanaz.ttf");
. . .
Loader.SetFontProvider(fontProvider);

```

MapFontToFile()

5 4

Scaleform

1.

/

2.

gfxexport

3.

Gfx::FontPackParams

4.

PC Xbox 360 PS3

Nintendo Wii

Gfx::FontProviderWin32 Gfx::FontProviderFT2

CPU

PSP PS2

gfxexport

CPU

Scaleform

5.1

Scaleform	Render::GlyphCacheConfig	Gfx::FontPackParams
	Renderer2D	

Scaleform

Gfx 4.0

Scaleform 3.x	4.0		Gfx 3.3	GfxLoader
	GfxFontCacheManager	Gfx 4.0	Renderer2D	
GlyphCacheConfig		Scaleform Gfx 4.0		5.2

Gfx::FontPackParams	'gfxexport'
---------------------	-------------

```

bool Done = false;

if (Font has Static Textures with Packed Glyphs)
{
    if (GlyphSize <
        FontPackParams.TextureConfig.NominalSize * MaxRasterScale)
    {
        Draw the Glyph as a Texture using Static Cache;
        Done = true;
    }
}
else if (Dynamic Cache is Enabled AND
        GlyphSize < GlyphCacheParams.MaxSlotHeight)
{
    Draw the Glyph as a Texture using Dynamic Cache;
    Done = True;
}

if (Not Done)
{
    Draw the Glyph as Vector Shape;
}

```

5.2

10000

```
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
renderer      Render::Renderer2D
```

HAL

LRU

```
Render::GlyphCacheParams gcparams;  
gcparams.TextureWidth    = 1024;  
gcparams.TextureHeight   = 1024;  
gcparams.MaxNumTextures  = 1;  
gcparams.MaxSlotHeight   = 48;  
gcparams.SlotPadding      = 2;  
gcparams.TexUpdWidth     = 256;  
gcparams.TexUpdHeight    = 512;
```

```
renderer->GetGlyphCacheConfig()->SetParams(gcparams);
```

TextureWidth, TextureHeight -

2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -

TexUpdWidth, TexUpdWidth – 256x256 128x128 256x512 (128K)

)

4KB 1MB 4 KB

256 256 4K

4K 16 100 256 1MB

4K – 256

MaxSlotHeight+2*SlotPadding 2-5 10 60-80%

500 2000

70-75

Flash

GFX, Dynamic cache	Readability	Animation
<i>Anti-alias for readability</i> <i>Anti-alias for animation</i>		

Scaleform

Flash

Scaleform
(O G C Q o g e)
Scaleform

- H, E, F, T, U, V, W, X, Z
- z, x, v, w, y

Scaleform

“Warning: Font 'Arial': No hinting chars (any of 'HEFTUVWXZ' and 'zxvwy'). Auto-Hinting is disabled.”

'Arial': ('HEFTUVWXZ' and 'zxvwy')

Flash
Zz” Basic Latin”

5.3 – gfxexport

gfxexport SWF GFX
SWF
DDS TGA –fc gfxexport

-fc	
-fcl <size>	25625625% Flash1024
-fcm	BooleanFontCompactor 10- 70% – (Hash) 12321664

5.4 - gfxexport

–fonts gfxexport
Scaleform Player Scaleform

gfxexport

-

-

-

CPU

Gfx::ImageCreator

Render::Renderer

'test.swf'

'test.gfx'

gfxexport -fonts -strip_font_shapes test.swf

-strip_font_shapes

Scaleform

gfxexport

gfxexport

-fonts	
-fns <size>	48 tri-linear mip-map
-fpp <n>	3
-fts <WxH>	256x256 '-fts 128' 128x128' -fts 512x128'
-fs	

SetGlyphCountLimit() 0

500

```
SetTextureConfig()  
    FontPackParams::TextureConfig fontPackConfig;  
    fontPackConfig.NominalSize    = 48;  
    fontPackConfig.PadPixels      = 3;  
    fontPackConfig.TextureWidth   = 1024;  
    fontPackConfig.TextureHeight = 1024;
```

NominalSize -

NominalSize

NominalSize

PadPixels - 1

TextureWidth, TextureHeight - 2

1) ;

```
2) Ptr<FontPackParams> packParams = *new FontPackParams();  
   Loader.SetFontPackParams(packParams);  
   ...  
   renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
3) Ptr<FontPackParams> packParams = *new FontPackParams();  
   Loader.SetFontPackParams(packParams);  
   Loader.GetFontPackParams()->SetGlyphCountLimit(500);  
   500
```

5.6

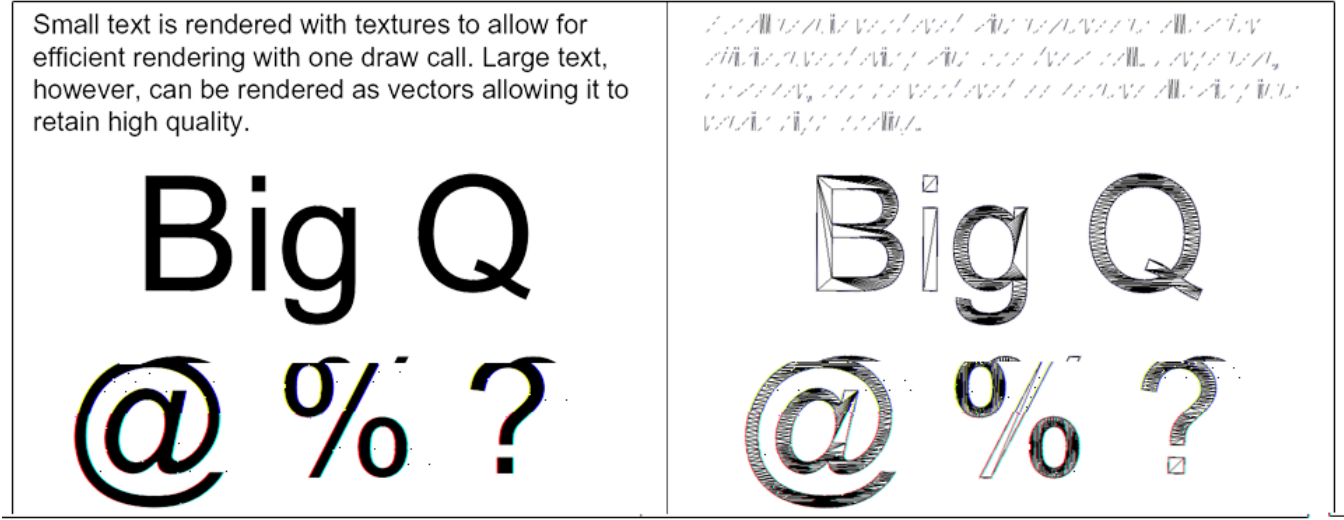
Flash

Scaleform

Scaleform

Scaleform Player

Ctrl+W



Render::GlyphCacheConfig::SetParams

MaxSlotHeight

SetMaxRasterScale

MaxRasterScale

1 1.25

GlyphCacheParams::MaxRasterScale

1.25

60

48

MaxRasterScale

gfxexport

GFx

-strip_font_shapes

6 5

Scaleform 2.x

Scaleform

Flash

Adobe Flash ,

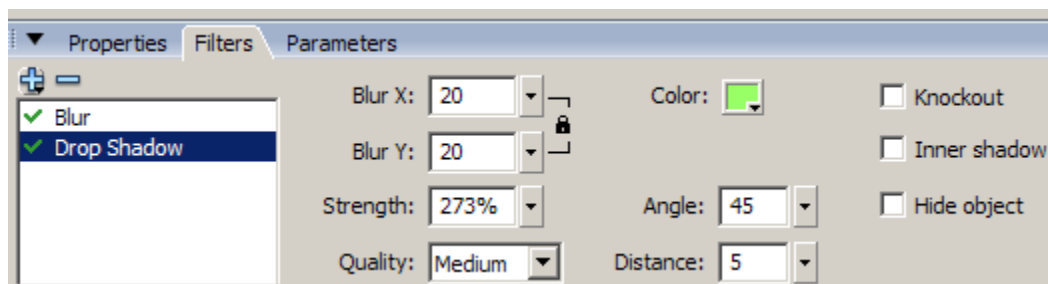
Flash

Scaleform

Scaleform

6.1

Flash Studio



Flash Scaleform
Flash

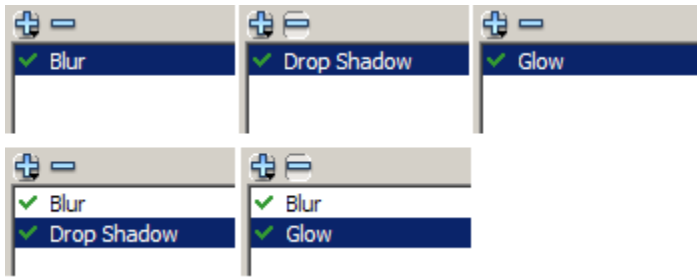
Scaleform

Scaleform

Flash

Scaleform

— — — — —



Scaleform

Flash

Knockout

Hide Object

Scaleform

Inner Shadow

Inner Glow

Scaleform

X Y

15.9

1590

Knockout

Flash

Scaleform

Flash

GfX

Text

Text

Knockout

6.2

Flash

3x3

3x3

;

Flash

Scaleform

Scaleform

Quality: Low

Scaleform
(sigma)

Scaleform

	Low quality	Medium quality	High quality
Flash	<i>Text</i>	<i>Text</i>	<i>Text</i>
GFx	<i>Text</i>	<i>Text</i>	<i>Text</i>

Flash

; Scaleform

Scaleform

<http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip-Smoothin.html>.

6.3

Flash

Scaleform

Scaleform

/

6.4 *ActionScript*

Scaleform				DropShadowFilter	BlurFilter	ColorMatrixFilter	BevelFilter
	AS2	AS3	/				Flash